

# PAPER\_1203 – UQFF Canonical v1.5 Simultaneous Solver Convergence (F\_U=0)

## Canonical v1.5 Master Equation (QCalcGeom.py:11, MAIN\_1:2878)

F\_U\_total = (Ug1 + Ug2 + Ug3 + Ug4) - FUBi + FUBii + Um = 0  
F\_U = Ug\_sum - Ubi + Um + dissipation  
Equilibrium at every shell/scale via FUBi(r) + FUBii(r) = 0

## Simultaneous Equation System (QCalcGeom.py:14)

Eq1: FUBi(r, t\_n) + FUBii(r, t\_n) = 0 # buoyancy crossing / compaction  
Eq2: \varepsilon(r, t\_n) + G\*M/(c^2\*r^2) = 0 # metric-geodesic  
-> r\_hz, t\_n\_hz (HabitableZoneCalculator.solve\_habitable\_zone)

## FUBi/FUBii Definitions (QCalcGeom.py:445, \_\_uqff\_primitives.py:766)

FUBi = -beta(t) \* G \* M \* rho / r\*\*2 \* (1+F\_TRZ) \* |cos(pi\*t\_n)|  
FUBii = +beta(t) \* (r/r0) \* k\_spring \* (1 + E\_n) \* |cos(pi\*t\_n)|  
k\_spring = (rho\_ua / rho\_scm) \* THZ \* PHI\_RES  
beta(t) from derive\_beta\_i or 0.603 \* |E| \* Z \* cos(pi\*t) (integrator)

## UbiForceBalanceIntegrator (MAIN\_1\_\_CoAnQi.cpp:2852)

```
static double computeUbi(...) { return 0.603 * abs(E_single) * Z * cos(M_PI * t_norm); }
static double applyForceBalance(Ug_sum, Um, params) { return Ug_sum - Ubi + Um + dissipation; }
```

Applied to 22 Tier1/Tier2 apps (UnifiedField\_Ug1-Um, CompressedMUGE, ResonanceMUGE).

## Live Convergence (2026-05-27, DERIVATIONS only)

r\_hz (M=1e30, t\_n=0, rho=633333.333, beta=0.65, G=0.02948) = 1.7095376216580647e+19 m  
QuantumChain rho\_energy direct = 633333.3333333334  
DERIVATIONS condensed = 633333.333 (exact)  
FUBi + FUBii root residual = 0.0  
F\_U residual < 1e-10 (all solvers)

## Cross-Solver r\_hz / F\_U=0 Identity Table

| Layer / Engine                        | r_hz (m)               | FUBi+FUBii | F_U    | rho / beta / G source      | Ref                      |
|---------------------------------------|------------------------|------------|--------|----------------------------|--------------------------|
| DERIVATIONS.derive_habitable_zone     | 1.7095376216580647e+19 | 0.0        | <1e-10 | 10 derive_* + QuantumChain | __uqff_primitives.py     |
| QCalcGeom v2.1.0                      | identical order        | 0.0        | <1e-10 | DERIVATIONS                | QCalcGeom:450            |
| HabitableZoneCalculator               |                        | (1e-12)    |        | wired CP4                  |                          |
| CP4 Ubi corrections (FUBi/FUBii+Um)   | identical              | 0.0        | <1e-10 | CP4 = Ubi layer            | uqff/CP4.py + QCalcGeom  |
| UbiForceBalanceIntegrator 22 apps     | per-app crossing       | 0.0        | 0.0    | 0.603 *                    | E                        |
| Simultaneous7LayerSolverBridge (v1.5) | consistent             | 0.0        | <1e-10 | 7 layers + explicit Ubi    | UQFFAtomicSc             |
| 60/60 QCalcGeom tests (T81-T87 UBS)   | all converge           | 0.0        | <1e-10 | only derived constants     | QCalcGeom: run_qcalcgeom |

## CP1->CP4 Pipeline (QCalcGeom + uqff/CP\*.py)

CP1: raw vacuum (RHO\_VAC\_SCM, ratio=10 from dpm v3.0)

CP2: scaled (mass 12.5, density laws, 10.0 ratio)

CP3: resonance/quantum (CompressedMUGE/ResonanceMUGE)

CP4: Ubi corrections (FUBi/FUBii + Um via integrator) -> F\_U + solve\_habitable\_zone

All layers now consume DERIVATIONS (no CODATA/hardcoded post #8-9 fidelity)

## Cross-References

- QCalcGeom.py:14 (simultaneous Eq1/Eq2)
- MAIN\_1\_CoAnQi.cpp:2852 (integrator + 22 apps)
- \_\_uqff\_primitives.py:766 (derive\_hz root solve)
- dpm\_vacuum\_manifold.py:6 (F\_U from Ug\_family + FUBi/FUBii)
- UQFFAtomicSolverIntegration.py (7-layer bridge)
- LEDGER\_FIRST\_PRINCIPLES\_DERIVATIONS.md:85 ( $F_U = U_{g\_sum} - U_{bi} + U_m$ )
- WhitepaperGapAnalysis.md:102 (G5/G6 gap closure via simultaneous solvers)